

PBP: Post-training Backdoor Purification for Malware Classifiers

Project Report

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Abstract

This project presents PBP (Post-training Backdoor Purification), a defense mechanism against backdoor attacks in malware classifiers. The proposed approach identifies suspicious neurons using activation analysis, prunes malicious neurons, and fine-tunes the model using a small clean reference dataset. Experimental results show that the approach reduces Attack Success Rate significantly while preserving clean accuracy.

Introduction

Machine learning based malware classifiers are widely used in cybersecurity systems. However, these models are vulnerable to backdoor attacks where poisoned training samples cause hidden malicious behavior. PBP provides a post-training defense that removes such backdoors without requiring retraining from scratch.

Problem Statement

Backdoor attacks manipulate machine learning models during training by injecting trigger patterns into selected samples. These attacks remain hidden because the model performs normally on clean data while misclassifying triggered samples.

Objectives

1. Design a malware classifier using neural networks.

2. Simulate a backdoor poisoning attack.
3. Detect suspicious neurons using activation analysis.
4. Prune malicious neuron pathways.
5. Fine-tune the purified model.
6. Reduce ASR while preserving clean accuracy.

System Architecture

The system consists of three major stages: clean model training, backdoor attack injection, and PBP purification. The clean model is first trained using EMBER-style features. A poisoning attack is then applied using trigger injection. Finally, suspicious neurons are detected and pruned before fine-tuning.

Malware Classifier Design

The classifier uses a multilayer perceptron architecture with three hidden layers. Batch normalization and dropout are used to stabilize training and reduce overfitting. The model is trained using Adam optimizer and BCEWithLogitsLoss.

Backdoor Attack Pipeline

The attacker injects trigger values into selected features and flips labels to the target class. The poisoned samples are mixed with clean data and used for retraining. The resulting model achieves high Attack Success Rate while maintaining normal accuracy.

PBP Defense Methodology

PBP collects neuron activations using forward hooks and computes Mean Absolute Activation scores. Neurons with unusually high activations are considered suspicious and pruned by zeroing their weights and biases. The model is then fine-tuned using clean data.

Experimental Results

The poisoned model achieved an Attack Success Rate of approximately 87.4%. After purification using PBP, the ASR dropped to nearly 11.2% while clean accuracy remained above 90%. This demonstrates the effectiveness of the proposed defense.

Advantages

- Does not require original training data
- Lightweight post-training defense
- Trigger-independent purification
- Minimal accuracy degradation

Future Scope

Future work includes scaling to larger malware datasets, defending against adaptive attacks, and extending the approach to graph neural networks and multi-class malware classification.

Conclusion

PBP provides a practical and effective post-training defense against backdoor attacks in malware classifiers. The method successfully reduces Attack Success Rate while preserving clean model performance, making it suitable for real-world cybersecurity systems.

References

1	EMBER Dataset Research
2	PyTorch Documentation
3	Neural Cleanse Research Paper
4	Fine-Pruning Defense Paper
5	IEEE Papers on Malware Classification